

Work Smarter, Not Harder

Company Overview: DataCEVA is a full-service data consulting firm and guide to data-driven excellence. We go beyond being a mere data consulting firm – we are your strategic partner in transforming businesses through insightful and decisive data solutions. We specialize in crafting dashboards that seamlessly transform into your daily navigation system. Informed decisions are the linchpin to success, and our mission is to provide tailored data services that align intimately with your unique business needs.

Project Overview: DataCEVA needs your help to build a customized AI-integrated internal operations system that allows us to stay organized and nimble as a small business. You will start by building a JIRA-like project management tool inside of a pre-existing communication and productivity platform. Then, you'll get the chance to connect that tool with an existing time tracking app, add a complex permissions structure, and integrate AI to give users the ability to make queries and autofill task tickets. You'll learn how to gather requirements from stakeholders, turn those requirements into user stories and acceptance criteria, interpret and integrate existing code with your own, test LLMs, and more.

Languages & Technology Stack:



Python



JavaScript



Bolt



Slack



OpenAI



DigitalOcean

Why Pick This Project: Instead of coding in a vacuum, you'll get to experience what it's like to work on a real world project from end-to-end. You'll start with the user interview process, create mockups, and then design and deploy a solution. By putting you in the driver's seat for the entire project, you get a chance to develop soft skills alongside your technical ones. Since you'll be integrated with DataCEVA engineers and building directly in our environment, you'll get to see the impact of your work from day one as this system will form the backbone of our operations for the foreseeable future.

Project Mentors



Dishant Rana | Founder & CEO | dishant@dataceva.com

Proud 2014 University of Wisconsin graduate with a degree in Economics and Statistics. I founded DataCEVA 5 years ago to help businesses take the next step towards becoming Data-Driven organizations. In my free time I enjoy playing golf (poorly) and rooting on any and all Wisconsin sports teams.



Abi Kasper | Project Manager, Engineering & Analytics |
abi@dataceva.com

Ohio State alumna (don't hold it against me) with a degree in Political Science and a minor in Statistics. I joined the DataCEVA team to standardize processes, get projects done more efficiently, and build client relationships. I have experience with everything from SQL to R to Tableau and consider myself a generalist in the analytics space. My expertise is in understanding how all the pieces of a project need to come together and in what order. When I'm not at my computer, I am usually playing in a volleyball tournament with my husband, coaching at F45, or curled up with a good book.

Scope of Work (SOW)

Project Title: Competitive Intelligence and Win/Loss Analysis Using AI

1. Project Overview

Flexera seeks to enhance its understanding of competitive dynamics and the reasons behind winning or losing deals. Currently, relevant data is fragmented across multiple systems including Salesforce (free-text fields), People AI (activity data), and 6sense (intent and engagement data). This project aims to consolidate, analyze, and visualize this data using AI models to generate actionable insights and support strategic decision-making.

2. Objectives

- Consolidate and structure competitive and deal outcome data from Salesforce, People AI, and 6sense.
- Apply AI models to extract patterns and insights from unstructured and structured data.
- Identify key drivers of deal wins and losses.
- Develop a dynamic dashboard to visualize insights and support ongoing analysis.
- Apply model to open pipeline opportunity data in Salesforce to apply the probability of the deal being won or lost. Build into dynamic dashboard that can be maintained by Flexera.

3. Scope of Work

Phase 1: Discovery & Data Assessment

- Review existing data sources and schemas.
- Identify key fields and data quality issues.
- Define success metrics and KPIs for win/loss analysis.

Phase 2: Data Integration & Processing

- Extract and normalize data from:
 - Salesforce (free-text fields, opportunity data)
 - People AI (email and activity logs)
 - 6sense (intent signals, keyword and competitor tracking)

- Clean and structure data for analysis.

Phase 3: AI Model Development

- Use NLP to analyze free-text fields and extract themes.
- Apply machine learning models to identify patterns in win/loss outcomes.
- Correlate activity and intent data with deal outcomes.

Phase 4: Dashboard Development

- Design and build an interactive dashboard (e.g., in Power BI).
- Include filters for time, region, product, competitor, etc.
- Provide drill-down capabilities for individual deals and trends.
- Apply model to open opportunity data in Salesforce to apply the probability of the deal being won or lost. Build into dynamic dashboard that can be maintained by Flexera.

Phase 5: Handover & Training

- Dashboard to be hosted and maintained by Flexera and deliver documentation and training for internal teams to ensure this happens.
- Provide recommendations for ongoing data governance and model retraining.

4. Deliverables

- Data integration pipeline and structured dataset
- AI-generated win/loss analysis report
- Interactive dashboard with real-time updates and model applying to open pipeline opportunities
- Documentation and training materials for ongoing use of dashboard post-project.
- Execute the project according to the scope and timeline.
- Maintain data confidentiality and security.
- Provide regular updates and status reports.



Project Ghost: Applying Captured 3D Human Data to Avatars to create a 3D Replay Engine on the Apple Vision Pro

This data is used by Holos to turn lived human expertise into structured 3D data to shape the AI models of tomorrow.

Project Mentors



Tyler Waite

Co-Founder, COO, VP of Product

✉ tyler@holos.io

Tyler leads Product by bridging the gap between engineering and business objectives. He holds a degree in Chinese Language and Literature from UW but comes from a design background. He was a part of a 3D printing startup while at UW, Spectrom, that sold to MakerBot in 2015. Tyler also leads deal flow within the Gov space, applying Holos' toolset to some of society's grand challenges.



Dan Borkhus

Co-Founder, CEO

✉ dan@holos.io

With a degree in Economics from UW, Dan manages business operations for Holos and was previously at Education Analytics in Madison. At Holos, he has brought in ~\$4M in military contracts and ~\$2m in private investment. He led the team through their participation in Techstars Space in 2020, and spearheaded Holos' strategic partnership with Scale AI to advance the integration of AR and AI, and manages relationships with Holos' investors, customers, and mentors.



Surya Narne

Unity Engineer

✉ surya@holos.io

Surya, a recent UW-Madison Computer Science graduate, is a Unity Engineer at Holos, involved in the development of software using the Apple Vision Pro. He gained experience in VR/AR game development during his time at UW, where he also participated in the Spring 2024 CS Capstone project. After working with Holos during this project, he later joined the company full-time.



Rea Mammen

Software Engineer

✉ rea@holos.io

Rea recently graduated from UW-Madison and participated in the capstone course during her last semester. She led her team's application on the Vision Pro, including over 6 distinct features, and can give you numerous tips on how to have a successful semester!



Ankit Kulshrestha

Unity Engineer

✉ ankit@holos.io

Ankit recently graduated from UW-Madison with a degree in Computer Science and Data Science. Having worked with Dan and Tyler in a previous cohort of the capstone class, he gained valuable VR/AR experience and later joined Holos in July as a Unity Engineer. He has been working on Unity and Apple Vision Pro development for the Holos app.



Saaketh Yerramsetti

Software Engineer

✉ saaketh@holos.io

Saaketh is a recent graduate from the UW-Madison with degrees in Computer Science and Data Science. As part of the Spring 2025 cohort, he led his team to a successful project and demos, and is eager to support and help wherever needed.

Ad Hoc Mentors



Yaj Shrotriya

visionOS Engineer

✉ yaj@holos.io



Jennifer Han

Augmented Reality

Product Designer

✉ jennifer@holos.io

Project Overview

At Holos, we've built a cutting-edge capture system on the Apple Vision Pro that records how experts perform physical tasks, from making espresso to maintaining aircraft. These recordings include **hand movements, body pose, narration, object tracking, and stereo video**.

This semester, you'll build a **replay engine** that uses this data to animate a 3D avatar demonstrating these tasks back to a learner. Think of it like "ghost mode" in a video game, but powered by real human motion data, running in an immersive training environment.

The goal is to take our pre-recorded datasets and **bring them to life** via avatar playback, synced narration, and clear motion fidelity. This will be a foundational shift for how users learn in Holos; by watching and mimicking the avatar.

This rich 3D behavioral data has immense value for foundational AI research at organizations like OpenAI, Meta, and Google. By capturing how humans move, interact with objects, and communicate in space, these datasets can help power the next generation of **multimodal AI models**. Such models will not only understand text, images, and audio, but also embodied human motion and skill execution, a critical step toward the development of **Artificial General Intelligence (AGI) and advanced robotics**.

Core Features:

- Import and parse a captured dataset (pose, hands, audio, metadata)
- Drive an avatar's animation using real-world joint and pose data
- Sync audio narration with avatar movement timeline
- Build a viewer in visionOS/RealityKit

Stretch Goals (choose one or more):

- Add callouts or “action markers” above key objects or moments
- Add motion smoothing to clean up animations
- Create a scene switcher to dynamically load and apply different datasets

Requirements

- **At least one MacBook (M1 or higher). These can be found at the library, too.**
- One Apple Vision Pro will be provided per team
- Builds to be pushed on TestFlight for collaborative testing with the Holos team



Skills & Technologies

Technology			Expertise
Coding	Prototyping	Process	Skills
		  	• Rapid prototyping + Iterative design thinking • Interest in avatars, XR, or motion capture • Model tuning/forking, ML pipelines

What We're Looking For in Team Members

We will ideally have two teams working in parallel and collaborating with Holos mentors. Each team will need a mix of technical depth, creative design skills, and curiosity about AI + XR.

Desired Roles & Skills

AI/ML Hobbyists

- Students with strong interest or experience in AI/ML, particularly motion capture, inference models, or multimodal data.
- Prior internship experience at organizations like **OpenAI, Google DeepMind, Meta AI, or other top research labs** would be a huge plus.
- Excited to explore how **3D human datasets** can be used to advance foundation models and embodied AI.

UI/UX & Design Minds

- At least one member on each team should have a strong background in **design and user experience**.
- Skills in **Figma** and **Adobe Creative Suite** are highly valuable.
- You will also be able to **tap into the expertise of our UI/UX designer, Jennifer Han** (jenniferhan.xyz), who brings deep experience in interaction design and spatial interfaces.

Engineering Roles

- Unity + C# developers ready to build on **Apple Vision Pro** (RealityKit, visionOS).
- Students interested in real-time animation, interaction design, and immersive prototyping.
- Comfort with importing, parsing, and working with real-world datasets (pose, hand, audio, metadata).

Qualities We Value Across Both Teams

- Creative problem-solving and willingness to experiment
- Rapid prototyping and iterative design thinking
- Collaboration and clear communication
- Interest in avatars, XR, AI, or motion capture
- Comfort working with **non-perfect, real-world datasets**

Why Pick This Project?

You should pick this project if you're passionate about the newest platforms and paradigms and have the desire to mold them into something that the world hasn't seen before. You'll be able to have a large amount of autonomy over the design of the solution developed but will also develop a strong sense of user-focused design that is so important for developing a valuable solution. You'll work at the intersection of XR (including the Apple Vision Pro), AI, and human learning, helping create a new interface for how people teach and learn physical tasks. This is a chance to ship something that feels like science fiction, and also sets the groundwork for real-world immersive training systems used by large enterprise organizations.

You'll also be contributing to a dataset and technology stack that has far-reaching implications: powering **next-generation multimodal AI models** and preparing for a future where **mass-market AI + AR glasses** from Apple, Google, Meta, and Samsung can capture and leverage 3D human knowledge and activity at scale.

You'll own your problem space, get mentored by a team who's built capture and training tools for the U.S. Air Force, and help define what instruction looks like in the age of avatars. This work has the potential to influence how leading AI companies train their foundation models, making 3D human data a new frontier for machine learning. By participating, you'll be contributing to breakthroughs that shape the future advancement of the **global AI market**.

We are a small but mighty team that gives people high ownership over their problem space, while also putting a huge emphasis on collaborating and staying organized as a team. We're looking for students who can do the same, and through this project, we hope you're able to develop the characteristics of a high-performing teammate. Join our team if you want to make a ding in the universe. **5 out of our 10 employees came directly out of this Capstone class and 7 out of our 10 employees are very recent college graduates.**



AI Powered Knowledge Content Creation

Project Mentors:

Nick Statz, Operations Lead, ops@qualtim.com

Nick graduated from UW-Madison in May of 2022. His capstone involved learning how to use Machine Learning to predict crop yields on a county-level basis across the Midwest.

Joe Schauer, IT Manager, jschauer@qualtim.com

Joe was involved in a UW-Madison Engineering EXPO award winning project during his time as a student, has a daughter that graduated from UW-Madison, and enjoys attending UW-Madison sporting events.

Tj Vita, Collaborative Intelligence Developer, tvita@qualtim.com

When Tj was younger he operated a BMX design agency where a passion for systems optimization spurred development of automated design systems and processes which facilitated significantly increased workloads.

Project Overview:

Turn raw data into real world impact by designing and prototyping a secure, on-prem AI system that:

- Finds the right information instantly through enterprise search & retrieval.
- Generates complete, standards-compliant draft reports from verified data.
- Formats outputs using Company templates - including figures, tables, and citations.
- Prepares drafts for expert review, cutting report creation time dramatically.

Skills & Technologies:

Gain skills in a project deployed to production involving:

- Prompt engineering
- ETL/ELT processes
- Data governance
- Exposure to vector databases and retrieval-augmented generation
- Implementation of an agentic framework with workflow orchestration
- API integrations
- Tools like LangChain for monitoring and auditing purposes

Why Pick This Project?

Qualtim is passionate about innovation and this project is at the forefront of it for the building construction industry! Work with a fun group of people to turn raw data into standardized reports that allow certified products to be used to build schools, hospitals, and homes nationwide.



COMPUTER SCIENCES
School of Computer, Data & Information Sciences

SCHNEIDER

Project Title: Enhancing Freight Search through Data-Driven Design

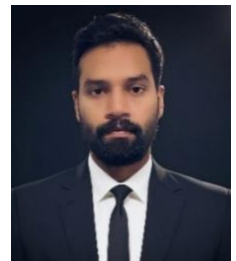
Project Mentors:

Gowtham Chowdary Sunkara – Data Scientist Sr

SunkaraG@schneider.com

Alma Mater: University of South Florida-Tampa, Florida

Bio: I'm a Senior Data Scientist at Schneider with close to six years of experience in the field of data science. My journey into data science began during my Master's in Industrial Engineering, where advanced analytics courses sparked a deep interest in data-driven problem solving. One of the key moments that solidified my interest in data science was a course project where I worked with class imbalance data and a capstone project involved building SQL databases for a community initiative. That experience challenged me to think critically about model performance and fairness, and it opened my eyes to the complexities and nuances of real-world datasets. At Schneider, I've worked on scalable machine learning models and data platforms that support strategic decision-making, with expertise spanning data engineering, predictive modeling, and business analytics.



College Memory: A standout moment from my college experience was receiving extra credit from my professor for the curiosity and depth of questions I brought to a project. That recognition reinforced my passion for data science and the value of staying inquisitive and engaged with complex problems.

Kate Romenesko – Solution Analyst Lead, Schneider FreightPower

Romeneskok1@schneider.com

Alma Mater: University of Wisconsin-Madison, Lakeland University

Bio: I have spent 20+ years at Schneider working in many roles, but always had a passion for tech. I started my journey working in driver pay, support, expenses and working my way from processing to auditing. I also lead many areas on process improvements and project work. I often found myself leading projects from the



business side like moving to the Oracle suites, Driver Pay upgrades, and 12 years ago I ventured into the world of tech as a solution analyst. Since then, I have led small to large scale projects to include most recently onboarding shippers and owner operators to the FreightPower tool.

College Memory: One of my favorite memories from college was, of course, attending the badger football games every week.

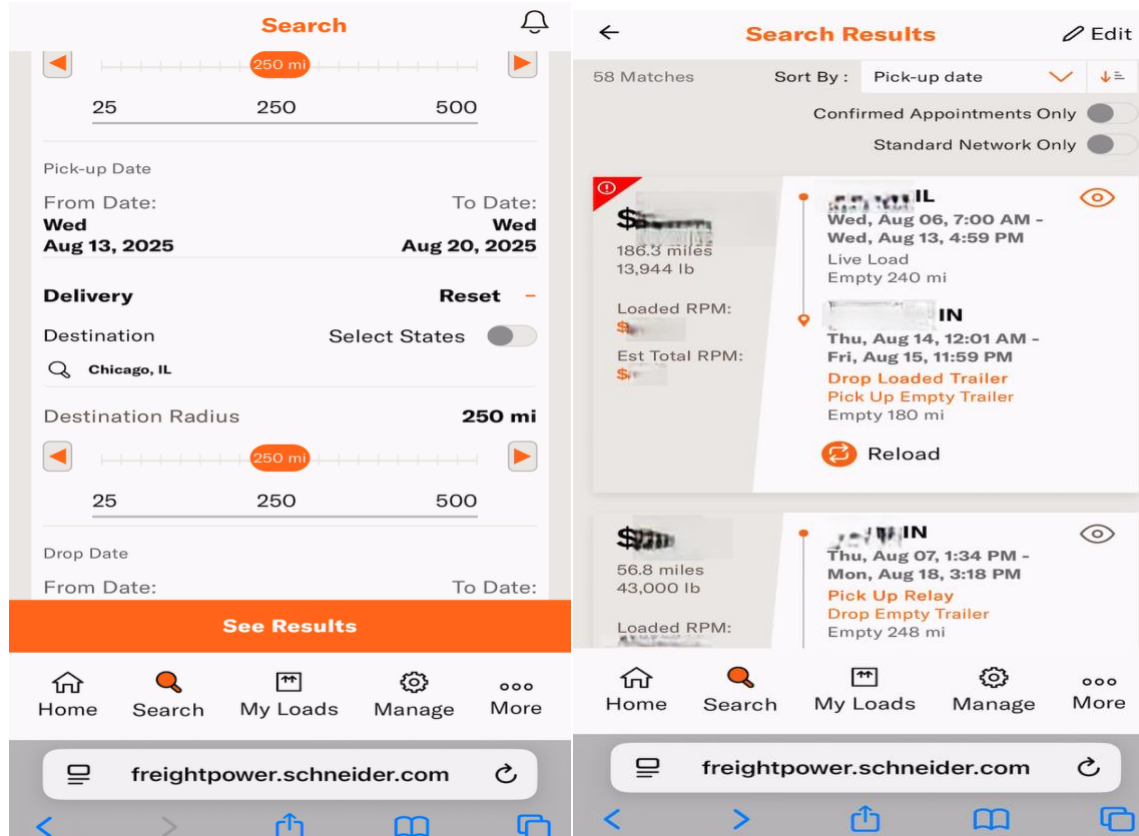
Project Overview:

Schneider FreightPower Owner Operator is a web app developed for drivers who have an owner-operator relationship with Schneider. The app allows users to search for and book available freight, as well as provide other updates required for their daily operations. The tool was recently launched, using an existing platform already serving over 17,000 monthly users outside of Schneider.

A key functionality for these users is our search engine for available freight; see the screenshot of the search tile below. Upon executing a search, users can view search results in a list format, ranked according to some criteria (e.g. pickup date, total rate); see screenshot of the search results displayed below.

We would like you to re-imagine the display of loads to streamline the booking experience for owner-operators; you will have recent digital click event data available to potentially help prioritize your design decisions. Some questions that may help direct your area of focus:

- Do users tend to scroll very far down to find the load they choose to book? If so, is there a way to more compactly display large sets of search results within a relatively small mobile screen area? Currently, 1-2 loads can be viewed at a time.
- Do users tend to re-sort their search results according to some criteria? If so, is there an opportunity to improve the default ranking of search results in some way that considers multiple criteria (pickup date, total rate, etc.) at once?
- Do users tend to click into load details before booking freight? If so, is there a way to efficiently display this load's information as part of the initial display, rather than via some sequence of click actions?
- Can we proactively display load recommendations even before users initiate a search, based on their historical search and booking behavior?



Skills & Technologies:

Programming Language: React, Python, Java

Tools: Python Libraries, ML/DL frameworks, Machine Learning models

Database: Anything open source, search-related data will be provided

Areas of expertise: UI/UX design and development, search algorithms, data engineering

Why Pick This Project?

Real world impact: Your solution could directly enhance the experience of Schneider's Owner Operators, as well as the over 17,000 monthly users of our legacy app for third-party freight carriers, making their daily operations smoother and more efficient.

Search engines and data engineering: Gain hands-on experience in designing and implementing robust search algorithms that handle large amounts of data, a highly sought-after skill in the tech industry.

UI/UX design: Practice designing a production level user interface that is intuitive, efficient, and user-friendly.



COMPUTER SCIENCES

School of Computer, Data & Information Sciences

Shopbop Social Recommendations



Project Mentors:



Ethan Falck (efalck@amazon.com)
Software Development Engineer II

Steven Domitrz (domitrzs@amazon.com)
Front End Engineer II

David Kassa (okdave@amazon.com)
Software Development Manager

A Favorite College Memory

Driving through the endless cornfields of Indiana.

Studying abroad in Rome and traveling around Europe.

Reverse engineering a binary file to create new private server accounts based on PayPal donations.

Project Overview:

Shopbop, an Amazon company, is a brand built on style, discovery, inspiration, and fun. To further engage our customers with these pillars, Shopbop is looking to explore social and interactive shopping experiences. We want to create something that will engage our customers with the constantly changing fashion industry. Furthermore, we would like to build brand-awareness and draw more customers to new products on our site. We have two projects, one for each team, to drive these new experiences:

1) Crowdsource the Best Clothes on Shopbop

Build a pairwise ranking system for comparing Shopbop products, similar to Flickchart for ranking movies (<https://www.flickchart.com/>).

The user is presented with items from the Shopbop catalog and chooses the one they prefer. Leaderboards are displayed of the most popular items by category.

For the final demonstration, present a link to a live system and allow the audience to rank a category. After data has been gathered, display the audience's favorite items.

Goals:

- Integrate with the Shopbop API to pull production information including images.
- Build a UI (web, native app, or both) to capture user choices and use a datastore to persist the collected data
- Display lists of the most popular clothing items by category
- Stretch Data Science goal - find users with similar preferences and offer recommendations
- Learn popular AWS technologies. This can include a web framework stack running on EC2 / RDS (e.g. Django, Ruby on Rails, various JS frameworks), a serverless approach (e.g. using API Gateway, Lambda, DynamoDB), or any other service you want to learn.

2) Find the Perfect Shopbop Outfit Match

Build a “Tinder”-like system for matching users with Shopbop outfits that they like.

The user creates a profile that stores their personalized data. They are then presented with either individual items or outfits comprised of multiple items from the Shopbop catalog and can swipe left or right to indicate they dislike or like that outfit, respectively.

For the final demonstration, provide a live screen-shared demo of a user swiping in the app and getting matches or even give the audience a link to the system such that they can use it themselves.

Goals:

- Integrate with the Shopbop API to pull production information including images.

- Build a UI (web, native app, or both) for the main application and use a datastore to persist the collected data
- Rank items in the back-end with a simple system such as ELO (used by Tinder)
- Provide a page for a user to see all of their “matches”
- Stretch Data Science goal – generate user style preferences based on shared characteristics of their matched outfits and intelligently recommend new outfits
- Stretch app goal – separate the outfit matching mode (swipe left or right on a full outfit) from the outfit generation mode (swipe left or right on individual clothing items and get “matching” outfits recommended)

Skills & Technologies:

You can develop with the technologies you choose. We can support students looking to develop their skills in UX, frontend, and backend engineering. We have expertise in...

- JavaScript & TypeScript
- React & React Native
- Swift
- Python
- Java
- Agile & Kanban development practice
- and the AWS Stack – EC2, S3, Lambda, RDS, Dynamo, API Gateway, and more!

If you choose a different tech stack, we'll be happy to support you but are less able to help.

Why Pick This Project?

Join the Shopbop team and transform the future of fashion with your coding skills! You'll get hands-on experience with real-world challenges and innovate using modern technology with industry experts.





AI-Powered Demo Automation for B2B Sales

PROJECT OVERVIEW

This project focuses on building go-to-market solutions to accelerate Strudel's sales cycle and customer discovery process.

Strudel is an AI-native startup that automates technical customer support. Students will partner with Strudel to build a comprehensive demo management and rapid prototyping platform, transforming the current manual, time-consuming demo customization process into an automated, intelligent workflow powered by AI.

The platform will feature:

- WYSIWYG content editors
- LLM-powered content generation
- Seamless API integrations with popular business tools (FreshDesk, Jira, Slack)

Students will gain hands-on experience in GTM (Go-To-Market) engineering—a rapidly emerging field where AI meets sales operations—while building tools that directly impact Strudel's ability to iterate quickly with prospects and customize demonstrations for different industries and use cases.

PROJECT MENTORS

Brian Kaufman | Chief Product Officer | brian.kaufman@strudelbots.com |



Brian is a 20-year veteran of software engineering and product management across HealthTech and AI, with experience at Epic, Huron, and XSell. Successful founder of Janus Health with a personal exit in 2023.

Favorite College Memory: *When I lived on Randall Street my Senior year, we had hot water included in our rent. Naturally, we bought a large inflatable pool and ran a bucket brigade to fill it with hot water from the sinks so we could have hot tub parties on our front lawn.*

Kristin Isaac | Chief Executive Officer | kristin.isaac@strudelbots.com



Kristin is CEO of Strudel, developing AI agent technology for enterprise technical support. She brings 20 years of go-to-market experience, including scaling businesses at LinkedIn and ESPN Digital, with deep expertise in enterprise partnerships.

She also serves as AI Advisor at the University of Wisconsin-Madison, where she connects industry partners with the university's cutting-edge AI research and expertise.

She holds an MBA from Columbia Business School and a BBA from UW-Madison.

Favorite College Memory: *I'm a recovering academic overachiever who took classes in CS, business, international development, politics, three languages (French, Spanish, Chinese), golf, AND ballroom dance. Yes, I was that student who treated the course catalog like a buffet—and I regret nothing! ;)*

SKILLS & TECHNOLOGIES

TECH STACK

- Python/FastAPI
- AWS
- REST APIs
- Frontend Framework (Student Choice for WYSIWYG Editor)

AI/ML

- ChatGPT/Claude/Gemini
- Prompt Engineering
- LLM Integration
- AI Workflow Automation
- Agentic Workflows

INTEGRATIONS

- FreshDesk API
- Jira API
- Slack API
- Webhook Management

ADDITIONAL SKILLS

- WYSIWYG Editor Development
- JSON/HTML Template Systems

- B2B SaaS Architecture
- Demo/Sales Engineering
- API Design & Development

WHY PICK THIS PROJECT?

Join a scrappy AI startup where your code will directly impact real customer conversations and sales outcomes, not just sit in a repository gathering dust. You'll get mentorship from actual founders who are actively scaling their company while building expertise in agentic workflows and GTM engineering that will set you apart in the job market.



Title: Scalable Systems for Civic Data Aggregation

Project Mentors:

Josh Cutler | CTO | josh.cutler@yardstik.com | [LinkedIn](#)

UW Alum, class of 2005, majored in Computer Science, Math and Political Science. I've worked all over the country in the tech world ranging from big tech (Microsoft and Optum) to being a multi-time founder at startups. I've been the CTO at Yardstik since January of 2024 where I lead the Product, Engineering and Customer teams.

Karen Nathan | VP Product | karen.nathan@yardstik.com | [LinkedIn](#)

I have 15+ years experience in Product with ranging from startups to multi-national companies, from business incorporation services and legal marketing to e-commerce (retail). I've been leading the Product team at Yardstik since April 2022.

Project Overview:

Who are we?

Yardstik is a Minneapolis-based background screening and employment risk startup on a mission to make trust and safety infrastructure accessible and measurably better.

What is the problem?

Most background check data originates from public sources—but ironically, it often reaches employers only after being funneled through expensive, outdated commercial aggregators. Think OFAC watchlists, state licensure databases, sex offender registries, and county-level court records. These sources are technically public, but practically buried under layers of manual processes, janky websites, and inconsistent formats.

This project will build a smarter pipeline: an intelligent system that gathers background screening data directly from the original source, bypassing the middlemen and unlocking a new level of speed, accuracy, and affordability in employment screening.

This is a real-world, full-stack systems challenge where civic data infrastructure meets machine learning and software engineering.

Skills & Technologies:

This project will both require and grow student skills in:

- Distributed systems & scalable processing — especially for high-volume web crawling and source syncing
- Database design & data modeling — including data normalization, expiration, and historical tracking
- Project & product management — scoping, planning, and iterative delivery in a real-world context
- Machine Learning & Natural Language Processing — for cleaning, structuring, and interpreting unstructured data
- Software Development Lifecycle (SDLC) fundamentals — version control, testing, CI/CD, monitoring, etc.

We are technology agnostic and excited to collaborate with student teams to choose the most appropriate tools, languages, and stacks for the job.

Why Pick This Project?

Background checks directly impact people's access to jobs and housing, and improving the speed and accuracy of that data can reduce delays, errors, and bias in life-changing decisions. By working directly with Yardstik's product and engineering leaders, students will gain real-world experience building production-grade systems with meaningful civic and social impact.

ARC Log Whisperer

Project Overview

GE HealthCare's Anesthesia and Respiratory Care (ARC) team supports clinicians with a suite of advanced medical devices designed for perioperative care. These devices generate vast amounts of internal log data, rich in insights but currently underutilized due to fragmented storage and unstructured formats. Engineers often face significant hurdles when trying to locate, access, and analyze logs to troubleshoot issues or identify performance trends. This results in missed opportunities to improve device reliability, efficiency, and patient outcomes.

The ARC Log Whisperer project aims to transform this challenge into an opportunity by developing a centralized, intelligent log analysis platform. This system will ingest raw text logs from various ARC devices, convert them into structured, searchable data, and provide intuitive tools for querying, visualizing, and alerting on key patterns and anomalies. The solution will include both backend services for data processing and storage, and frontend tools such as a web interface for user interaction.

By enabling engineers to quickly answer questions like “Have similar alarms occurred recently?” or “Are there unusual loading patterns across devices?”, the ARC Log Whisperer will empower faster, more informed decision-making. The project will also explore the use of AI to support natural language queries and automated anomaly detection, paving the way for smarter diagnostics and proactive system monitoring.

You should Pick this Project!

This project offers a unique opportunity to make a meaningful impact on how engineers at GE HealthCare interact with and learn from the vast log data generated by anesthesia and respiratory care devices. If you're passionate about solving real-world problems with data engineering, backend development, and intuitive user interfaces, this is the project for you.

The ARC Log Whisperer project is ideal for students who are curious, motivated, and eager to build something that matters. Your work will lay the foundation for future tools that enhance patient safety and operational efficiency. Plus, you'll be part of a team that values creativity, initiative, and continuous learning, all while collaborating with experienced professionals in a high-impact healthcare environment.



Your Mentors

Dave Smith



Chief Engineer, Anesthesia and Respiratory Care (University of Pittsburgh BS ME '92) David.Smith12345@gehealthcare.com

A passionate innovator, mentor, and engineer with over two decades of experience driving breakthrough medical device product development. I thrive on transforming complex challenges into market-ready solutions. I bring a strategic mindset, a love for mentoring, and a hands-on approach to innovation—whether it's guiding startups or leading global teams through high-impact R&D. I believe in the power of always asking, “What’s next?” Looking forward to collaborating and creating what’s next together!

Paul Hooper



Staff Software Eng., Anesthesia and Respiratory Care (UC Berkely BS Engineering Physics) Paul.S.Hooper@gehealthcare.com

I greatly enjoy creating things and making things work, whether it be my car, working with wood, plumbing, tiling, electrical, a home appliance, a home security or lighting system, developing a desktop application, working on an embedded system, or creating services in the cloud to serve our needs. It used to be really tedious to track down technical information to understand how components and services worked in order to leverage/repair/modify them, but as the internet has grown and now especially with AI, we have an abundance of information to work with, and a great many tools to choose from, all which make the creative process even more interesting and enjoyable. The challenge for us will be to determine which technologies best fit our needs, and as we do this, a valuable by-product is that we will become familiar with technologies that may not fit well our immediate needs but may be useful to us in the future. What will be especially exciting about the ARC Log Whisperer project is that we will be working together at the ground level to establish the foundation for future development. We will not be saddled with pre-chosen technologies/tools and will have great freedom to explore and to create an awesome log analysis tool for GE Healthcare. Maybe you will want to rejoin us in the future to continue the work 😊.

Gene Chyou



Verification Team Lead, Anesthesia and Respiratory Care (UW-Madison BS Biological Systems Eng '05) genechyou@gehealthcare.com

Former Bucky Badger mascot turned seasoned medical device engineer and program manager. What I find most rewarding is mentoring others and helping them grow in their careers—a passion I get to pursue as GE HealthCare’s Campus Recruiting Leader for UW.

For this capstone project, I hope to offer a bit of everything: technical guidance, help with scoping and timelines, and conversations about your future career paths. I’m excited to collaborate with you all. . . and GO RED!

Macy Frank



Systems Eng., Anesthesia and Respiratory Care (UW Madison BME 2023) macy.frank1@gehealthcare.com

As a recent graduate from UW-Madison, I am excited to be back working with current students on this capstone project. Working at GE Healthcare allows me to get creative and hands-on with medical devices that are actively being used on real patients in the field. It's incredibly rewarding to contribute to innovations that make a difference in the medical field and have fun doing it. Looking forward to seeing what you create and helping you bring it to life!

Just in Time Compute Provisioning

Project Overview

Today, GE HealthCare's Portrait Mobile wireless patient monitoring ecosystem requires an extremely manual installation process of the compute platform. This process requires that the software be loaded directly onto servers during manufacturing or that field engineers manually upload and install software using USB sticks. This process is insecure, lengthy, and prone to errors.

GE HealthCare has been collaborating with other technology partners such as Intel, Lenovo, and Spectro Cloud to create an open source technology that enables true last minute provisioning. This would allow an "off the shelf" device to be provisioned directly at the customer site simply by turning on the device and configuring it via a phone application, much like smart home devices are set up today.

For this project, you will be creating a mobile phone application that will interact directly with the "off the shelf" device as well as the cloud backend in order to perform true, last minute provisioning of the device. This project will include working with multiple technology partners as well as both front and backend development.



You should Pick this Project!

Imagine a scenario where you can provision a device that is running life critical patient services as easily as you provision a smart lightbulb in your home. In this project, you will be making that a reality! You will be performing full stack development in order to build a mobile phone application that enables easy configuration of the compute. This project will include full stack development as well as exposure to databases, API creation, AI, geo-location, cybersecurity, UEFI, and much more. Most excitingly, this software will enable easy and secure configuration of compute that runs life critical patient monitoring services!



Your Mentors

Matt Pekarske



Principal Engineer, Monitoring Solutions (UW – Madison BS ECE '00) matthew.pekarske@gehealthcare.com

I've spent my entire career in connectivity and communications systems ranging from cellular test equipment to satellite TV, and since 2006, connected medical devices. My time at GE HealthCare has been by far the most rewarding work in my career due to its direct impact to the quality of patient care through those installing, using and servicing our products. I graduated from UW Madison and I'm excited to work with you!

Kate Hall



Software Engineer, Monitoring Solutions (Marquette BS BME '22 & UPenn MCIT expected '26) kate.hall@gehealthcare.com

I graduated from Marquette University in May 2022 with a degree in Biomedical Engineering before joining GE HealthCare's Edison Engineering Development Program. During my time at GE HealthCare, I have had opportunities to work directly with wireless communication networks, software engineering and agile, and data engineering. I have worked directly on our compute platform, helping to improve serviceability and the installation process, and enjoy helping create solutions that can have meaningful impacts on the lives of our customers. As a young engineer, I look forward to working with UW Madison students as they continue their engineering education.

Matt Grubis



Chief Engineer, Monitoring Solutions (UW – Madison BS ECE & CS '97, Marquette MS EE '01), matthew.grubis@gehealthcare.com

A tinker-er, explorer, inventor, mountain biker and a proud UW-Madison Engineering and Computer Science Graduate. Been working at GE HealthCare a long time; and it's the cool tech, the global reach, our super talented engineering team, and I love that I'm working on products that save lives that keeps me here. My family is a "Badger family" with both my sons either graduating from or currently enrolled in UW-Madison Engineering! Looking forward to meeting you and working with you, I think you'll love the things we do.

AI-Powered Physical Feedback Coach for Public Speaking

Project Description

Develop an application for Snapdragon PCs that analyzes video of a speaker to provide **quantitative, actionable feedback** on physical aspects of public speaking using AI models.

- Skills Needed:
 - **Programming:** Python (intermediate to advanced)
 - **AI Integration:** Experience with integrating AI/ML models into applications
 - **Edge AI:** Willingness to learn Qualcomm's AI SDKs (QAI RT, ONNX Runtime, LiteRT)
 - **Computer Vision:** Familiarity with pose estimation, facial landmark detection, and gaze detection

Phases

- **Research & Planning**
 - Study existing speaker coaching tools
 - Define physical traits to evaluate (posture, gestures, gaze, etc.)
 - Establish scoring metrics and feedback criteria
- **Model Selection**
 - Explore Qualcomm AI Hub for:
 - Pose estimation models
 - Facial landmark detection models
 - Gaze detection models
 - Evaluate models for accuracy, performance, and compatibility
- **Application Design**
 - Decide between **real-time** vs. **pre-recorded** video analysis
 - Design UI for video input, feedback visualization, and scoring
 - Plan architecture for model inference and feedback generation
- **Implementation**
 - Integrate selected models using Qualcomm SDKs
 - Build video processing pipeline
 - Implement scoring and feedback logic
 - Optimize for on-device inference (NPU if applicable)
- **Testing & Evaluation**
 - Collect or record public speaking videos
 - Compare AI feedback with human evaluations
 - Refine scoring and feedback algorithms
- **Documentation & Presentation**

- Technical documentation (architecture, model usage, deployment)
- User guide for application
- Final demo and presentation

Anticipated Technical Challenges

- On-device model deployment and optimization
- Real-time vs. batch processing trade-offs
- Synchronizing outputs from multiple models
- Gaze detection accuracy and calibration

Benefits to Students:

- Hands-on experience with **edge AI deployment**
- Exposure to **Qualcomm's AI toolchain and hardware**
- Skills in **computer vision, model evaluation, and application development**
- Potential for **model fine-tuning and custom deployment**
- Portfolio-ready project with real-world relevance

Skills Required

- Python, experience with integrating AI into purpose-built applications
- Willingness to learn Qualcomm's AI SDKs, with a focus on pose estimation models

Hardware

- Snapdragon PC (e.g., **Snapdragon X Elite** or **X Plus**)
- Webcam or video input for speaker footage

Deliverables

- **Project Proposal Document**
 - Problem statement and motivation
 - Scope and objectives
 - Initial research on existing tools and gaps
 - Proposed approach and evaluation criteria
- **Model Evaluation Report**
 - Comparison of pose, facial landmark, and gaze detection models
 - Justification for selected models (accuracy, performance, compatibility)
 - Notes on any BYOM (Bring Your Own Model) usage
- **System Architecture Diagram**
 - Overview of application components
 - Data flow from video input to feedback output
 - Real-time vs. pre-recorded processing decision

- **Working Application**
 - Runs on Snapdragon PC (X Elite or X Plus)
 - Accepts video input (live or recorded)
 - Performs:
 - Pose estimation
 - Facial landmark detection
 - Gaze detection
 - Outputs:
 - Quantitative scores for physical traits
 - Visual feedback (e.g., overlays, heatmaps)
 - Actionable suggestions for improvement
- **Technical Documentation**
 - Setup instructions
 - Model integration and deployment steps
 - Qualcomm SDK usage (QAI RT, ONNX Runtime, LiteRT, etc.)
 - Troubleshooting guide
- **User Guide**
 - How to use the application
 - How to interpret feedback
 - Example use cases
- **Evaluation Dataset**
 - Collection of public speaking videos (can include student recordings)
 - Used to test and validate feedback accuracy
- **Performance & Accuracy Report**
 - Evaluation of feedback quality (quantitative + qualitative)
 - Comparison with human assessments (if applicable)
 - Limitations and areas for improvement
- **Final Presentation**
 - Live demo of the application
 - Summary of technical challenges and solutions
 - Key learnings and future directions

Project Title:

Electronic Patient Reported Outcomes (ePRO) data capture

Project Overview:

RegenMed provides a software platform to collect clinical treatment information from both clinicians and patients. RegenMed's platform has two data collection modules, one for the investigator/clinicians and one for the patient. The data collected from the patient is often referred to as ePRO (electronic Patient Reported Outcomes). This data measures outcome criteria that patients will report after a series of medical procedures. The module for the patient consists of both a Web application and a mobile application. Patients do not always complete their surveys in a timely manner. We desire a new or enhanced patient collection software module. Enhancements would include updating the User Experience and adding a type of gamification or reward system.

Skills & Technologies:

Amazon Cloud: ECS, EC2, Lamda, Cognito, S3
.Net C#, React JS
MS SQL, Dynamo DB
Atlassian products (JIRA, Confluence, and BitBucket)

Why Pick This Project?

Utilize your frontend UI skills, your web and mobile programming skills, along with any gamification skills to make answering surveys more enjoyable. Healthcare Spending is at 5 trillion dollars in the US alone. Much spending is directed towards reactive treatments that have very little research or knowledge about that treatment's effectiveness. RegenMed is attempting to 'democratize' the availability of this treatment effectiveness data. We desire clinicians and patients to share the rewards that can be gained from making this data widely available to the healthcare ecosystem. Visit us at www.rgnmed.com to learn more. Join our project and start your career in this ever-evolving health care field.





UW CS639 Capstone Fall 2025

Project Mentors:

[Dolph Courchaine](#) | Chief Information & Technology Officer | dcourchaine@rgnmed.com



Dolph has been in the Software Engineering and the HealthCare fields his whole 40+ year career. He started Computer Programming while in high school and graduated from the University of Wisconsin – Madison in 1984 with a Bachelor of Science and a major in Computer Science. While obtaining his degree he became the first Computer Programmer for the Madison based Staff Model HMO, Group Health Cooperative of South-Central Wisconsin, which owned and operated clinics while taking insurance risk. This start led him to becoming a deep domain expert at the intersection of Health Care Insurance with Health Care Delivery.

[Will Graupmann](#) | Software Engineer | wgraupmann@rgnmed.com



Will graduated from the University of Wisconsin – Madison in December 2024 with a Bachelor of Science in Computer Science. He joined RegenMed in May 2024, where he has been analyzing the existing codebase and AWS architecture to improve efficiency, scalability, and the overall user experience of RegenMed Admins and Investigators.

[Laura Prey](#) | Board Member & Executive Consultant | lm prey@gmail.com



Ms. Prey brings to the RegenMed board more than three decades of senior-level IT experience in the healthcare, insurance and financial industries. As a proven visionary leader in Fortune 500 as well as small to mid-size companies, Laura established an enviable track record of accelerating the delivery of innovative technology solutions despite complex business and technical challenges. Ms. Prey received a Bachelor of Business Administration degree in Management Information Systems and an MBA from the University of Wisconsin Madison.



COMPUTER SCIENCES

School of Computer, Data & Information Sciences



Project Title: Build AI That Keeps Planes in the Air

Project Mentors:

Mark Mann – Head of Technology

mmann@smartcert.tech

Favorite college memory: building a race car robot armed with sensors to navigate hazards. This was an early dive into applied technology long before AI as we know it today.

Matthew Waller – Director of Engineering

mwaller@smartcert.tech

Favorite college memory: watching a passionate physics professor demonstrate velocity and vectors by kicking chairs. This was so memorable that students remixed his lectures into a techno track.

Project Overview:

AI is everywhere, from phones to productivity tools but embedding AI into critical, real-world applications comes with challenges: data security, cost, liability, and user trust.

SmartCert is transforming how companies handle supply chain documentation, a space where errors can have serious business consequences.



Build AI That Keeps Planes in the Air

This isn't a classroom exercise. One missing or incorrect certification can:

- Put unverified parts into airplanes, rockets, or medical devices
- Halt production lines
- Delay multimillion-dollar shipments
- Trigger costly audits and fines
- Damage supplier relationships

💡 **Your Challenge:**

Design AI that guides non-technical users through complex supply chain documentation, catching mistakes before they ground flights, delay launches, or stall production.

This is your chance to work on **real-world, high-stakes AI** that companies will actually use.

You will explore:

- Secure data/document access
- Extracting and analyzing critical information
- Using AI to research relevant regulations and standards
- Identifying mismatches or compliance risks
- Designing a user experience so intuitive “even your grandmother could use it”

You'll touch multiple layers of technology: from front-end interfaces and server-side logic to AI prompting, modern databases, and automated cloud deployments.

Skills & Technologies:

- React
- .NET / C#
- AI Prompting & Workflow Design
- JSON & REST APIs
- MongoDB
- Cloud-hosted CI/CD

Why Pick This Project?

- **Proven Mentorship:** Learn directly from startup leaders with a track record of building and scaling successful global software products still in use today.
- **Full-Stack Experience:** Gain exposure to every stage of the product lifecycle or dive deep into the area that excites you most.
- **Real-World Impact:** Your work won't stay in the classroom. It will directly contribute to solving real industry problems in aerospace and advanced manufacturing.
- **Collaborative Growth:** Instead of handing out assignments, we'll coach you on how real teams overcome challenges, make trade-offs, and deliver products that matter.
- **Résumé-Ready:** Leave with a portfolio piece that demonstrates your ability to work on enterprise-level problems with cutting-edge AI.

Our goal is simple: to make this project fun, challenging, and meaningful while giving you an experience that sets you apart in your career journey.